

ASTR 210 Fall 2025 — Homework set 11
50 points plus optional extra credit
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Solutions need to show all important intermediate calculations, diagrams, and explanations in addition to the final answer to receive full credit. Please check the legibility of homework solutions uploaded to the class website as we cannot grade or give credit for illegible or unreadable work. Units do not need to be specified if an expression or equation is requested; for numerical calculations please use the specified units for the answer, or if not specified, SI or cgs units. Please use Table A.1 for the value of physical constants and Table A.2 for the value of astronomical constants, unless otherwise specified in the problem. Please use the correct number of significant figures.

11.1 Spiral galaxies – Tracing the Light

Most of the luminosity of spiral galaxies such as the Milky Way comes from their galactic disks. The luminosity per unit area of such disks, $\Sigma(r)$, is typically very well approximated by the functional form:

$$\Sigma(r) = \Sigma_0 \exp(-r/h_r).$$

In this expression Σ_0 is the surface brightness of the disk at the center of the galaxy and h_r is a characteristic scale length for the radial fall-off of surface brightness.

- (a) Integrate this surface brightness distribution over radius to show that the luminosity of the galactic disk within radius r is

$$L(r) \equiv \int_0^r 2\pi s \Sigma(s) ds = 2\pi h_r^2 \Sigma_0 \left[1 - \left(1 + \frac{r}{h_r} \right) \exp\left(\frac{-r}{h_r}\right) \right].$$

Hint: $\int x e^{-x} dx = -e^{-x}(x + 1) + \text{constant}$. [3 pts]

- (b) In the Milky Way, our position in the disk, and obscuration by dust, prevent us from measuring Σ_0 directly. However, we can determine the local surface brightness of the disk, $\Sigma(R_0)$, at our distance R_0 . We can also measure the scale length of the disk. We find $\Sigma(R_0) = 36 L_\odot \text{ pc}^{-2}$, $h_r = 4 \text{ kpc}$, and $R_0 = 8.5 \text{ kpc}$. What central surface brightness, Σ_0 (in $L_\odot \text{ pc}^{-2}$), is implied for the Milky Way by these values? [4 pts]
- (c) What is the corresponding total luminosity $L(r \rightarrow \infty)$, in $L_\odot$, of the galactic disk? [2 pts]
- (d) Stellar populations are frequently characterized by their “mass-to-light ratio,” which is just their mass (in $M_\odot$) divided by their luminosity (in $L_\odot$). Our Sun has a mass-to-light ratio of 1, clearly. If the Milky Way’s mass consisted mostly of stars, with a typical mass-to-light ratio of 3, what would we infer for the luminous mass of the MW? [2 pts]
- (e) What fraction of the galactic disk luminosity arises interior to the orbit of the Sun? Hint – that fraction can be written as $L(R_0)/L(\infty)$. [2 pts]

Aside: The central bulge and halo of the Milky Way contribute an additional $\sim 20\%$ to the total luminosity of the Galaxy, and nearly all of that comes from galactocentric distances $r < R_0$.

11.2 Spiral galaxies – Tracing the Matter

In HW 10 we used the circular rotation speed of the MW galaxy to estimate its mass. We used a rotation velocity v_c or $\Theta = 235 \text{ km s}^{-1}$ at a radius of 17 kpc ($2R_0$), and you should have found a mass of about $2 \times 10^{11} M_\odot$. In this problem we’ll look at some alternate measurements that we can also use to estimate the mass of the Galaxy.

- (a) The highest-velocity stars in the solar neighborhood have speeds of order 500 km s^{-1} in the galactic reference frame. We know that they are bound to the Galaxy (they are not escaping) because we see the same number of them moving towards the galactic center as away from it. Thus, 500 km s^{-1} is a lower limit to the escape velocity from the Milky Way at the Sun's orbit ($r = R_0$).¹ Use that escape velocity to estimate the minimum mass of the Milky Way (in $M_\odot$). [4 pts]
- (b) After correcting for the Sun's orbital motion about the galactic center, one finds that the Andromeda Galaxy (M31) is actually approaching the Milky Way with radial velocity $v_r(\text{M31}) = -119 \text{ km s}^{-1}$. This measurement implies that the Milky Way plus Andromeda form a gravitationally bound system. In other words, their combined mass is large enough that their mutual gravitational attraction has overcome the expansion of the universe (in the time elapsed since the Big Bang). We can crudely estimate the total mass needed to accomplish this feat as follows.
- Estimate the distance to the Andromeda Galaxy. Consider a Cepheid variable star: it has a period-averaged apparent visual magnitude of $m_V = 18.9$ and its measured pulsation period is $P = 31.47 \text{ d}$. Assuming the Cepheid period-luminosity relation

$$M_V = -2.76 \log\left(\frac{P}{10 \text{ d}}\right) - 4.16,$$

compute the distance to M31 assuming no interstellar extinction along the line of sight. Quote your distance value in Mpc. [4 pts]

- Assuming a Hubble constant $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$, and the distance you determined above, compute the recession velocity Andromeda ought to have based on Hubble's law. [2 pts]
- Recall that an acceleration is $\Delta v / \Delta t$. If M31's velocity changed from the value in part (ii) above to the current value of -119 km s^{-1} over 14 Gyr, what average acceleration (in m s^{-2}) is implied? [3 pts]
- Use Newton's law of gravity to write down a simple expression for the gravitational force of M31 on the Milky Way (and vice versa). What is the mass of our Galaxy if this force supplies the average acceleration calculated in the previous part? You may neglect the fact that the distance from the Milky Way to Andromeda has clearly varied with time, as this is just an order-of-magnitude estimate. Also assume that $M_{\text{M31}} \approx 2M_{\text{MW}}$. [4 pts]

11.3 Black hole masses

Figure 20.9 in the text shows the rotation curve of material in the very center of the galaxy M87. This rotation curve provided some of the best evidence for a black hole in M87 — that is, until the Event Horizon Telescope imaged the “shadow” resulting from the event horizon. The rotation curve is approximately Keplerian at radii larger than $0.1''$.

Use these data to estimate the mass enclosed within a radius of $0.1''$ in the center of M87. Since the data are still in observed units, some extra conversion steps will be necessary.

- Find the linear radius (units of km or pc) corresponding to $0.1''$ at the distance of M87, which is approximately 16.8 Mpc. [2 pts]

¹The mass one derives for the Milky Way using this escape velocity argument depends on how that mass is distributed with radius, but one can show that the minimum mass follows from assuming that all of the mass of the Milky Way is concentrated at its center.

- (b) From the figure, read off the value of the velocities at slit positions of $+0.1''$ and $-0.1''$. (Sanity check: the velocity at $+0.1''$ is somewhere between 600 and 700 km s^{-1} . You should estimate it a little more accurately than that.) The circular velocity value that you want is half of the difference between those two velocities. [2 pts]
- (c) Then estimate the mass of the black hole. [3 pts]

11.4 Redshifts and the Hubble Law

Suppose you have measured the wavelength of the $\text{H}\beta$ line in an external galaxy; the rest wavelength of $\text{H}\beta$ is 486.135 nm , but you find an observed wavelength of 497.97 nm .

- (a) Calculate the redshift z of the galaxy. [3 pts]
- (b) In the absence of interstellar extinction (an assumption that is more or less OK for sources far from the plane of the Milky Way), the usual distance modulus relation is $m - M = 5 \log(d/10 \text{ pc})$. Rewrite this relation using the Hubble Law and redshift to show that $m - M = 5 \log z + 43.2$. For this rewriting, you should use the nonrelativistic limit ($z \ll 1$) and a Hubble constant of $70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. [5 pts]
- (c) The galaxy has an apparent visual magnitude of $m_V = 12.7$. Using your previous results, estimate the absolute visual magnitude M_V for this galaxy. [3 pts]
- (d) Based on absolute magnitude alone, is this galaxy more likely to be a giant elliptical or a bright spiral? [2 pts]

EXTRA CREDIT

11.5 Consider the following poem. Below you'll find some questions about it.

Some time ago my late Papa
 Acquired a spiral nebula.
 He bought it with a guarantee
 Of content and stability.
 What was his undisguised chagrin
 To find his purchase on the spin,
 Receding from his call or beck
 At several million miles per sec,
 And not, according to his friends,
 A likely source of dividends.
 Justly incensed at such a tort
 He hauled the vendor into court,
 Taking his stand on Section 3
 Of Bailey "Sale of Nebulae."
 Contra was cited Volume 4
 Of Eggleston's "Galactic Law"
 That most instructive little tome
 That lies uncut in every home.
 "Cease," said the sage, "your quarrel base:
 Lift up your eyes to Outer Space.

See where the nebulae like buns,
 Encurranted with infant suns,
 Shimmer in incandescent spray
 Millions of miles and years away.
 Think that, provided you will wait,
 Your nebula is Real Estate,
 Sure to provide you wealth and bliss
 Beyond the dreams of avarice.
 Watch as the rolling aeons pass
 New worlds emerging from the gas:
 Watch as the brightness slowly clots
 To eligible building lots.
 What matters a depleted purse
 To owners of a Universe?"
 My father lost the case and died:
 I watch my nebula with pride
 But yearly with decreasing hope
 I buy a larger telescope.

Originally published in *The Observatory* 65, 88 (1943). Reprinted from A RANDOM WALK IN SCIENCE, an anthology compiled by R. L. Weber and published by Institute of Physics, 1973; also reprinted by V. Trimble in "Star Formation Near and Far" (Eds. Holt and Mundy, 1997).

- (a) The author states that the galaxy in question is a spiral galaxy. What other pieces of evidence are provided about the galaxy's type, and how do they support that classification? [4 pts]
- (b) The distance to the galaxy is described as "millions of miles." Is that value consistent with its recession velocity? Of course we don't have very precise numbers, but you can make an order of magnitude estimate to see whether the values are plausibly consistent or whether the author is leaning heavily on poetic license. [4 pts]
- (c) Based on the historical material in chapter 20 and the lecture slides, speculate on the rough timeframe when the fictitious father would have purchased the nebula. Explain your reasoning. It's OK to cite a range of years or decades – we can't do any better than that. [2 pts]